IÉSEG SCHOOL OF MANAGEMENT

Master in Big Data Analytics for Business

PREDICTING SME BANKRUPTCY IN FRANCE:

A Machine Learning approach to Macro-Micro dynamics in a High-Interest Rate regime

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ABSTRACT

French SMEs are facing a major shift, moving from years of easy money to a period of high interest rates. This study builds a precise bankruptcy prediction model that connects firm-level financial weakness with broader economic shocks. Using data from about 268,000 French SMEs in the Orbis database, the research trains models on the stable 2016–2022 period and tests them on the more turbulent 2023–2024 period.

The experimental findings indicate that the Baseline XGBoost model is the best-performing algorithm, achieving an Out-of-Time (OOT) AUC of 0.68. A key theoretical contribution is the discovery of the "Tuning paradox," where models finely tuned for conditions before 2022 experienced performance drops (AUC 0.57), emphasizing the risks of algorithmic overfitting amidst regime shifts. The study also decodes the "black-box" nature of the ensemble learners by integrating SHAP (SHapley Additive ExPlanations), revealing that leverage—proxied by a median 0.73 Debt Ratio—acts as a dormant risk that is only "activated" when combined with rising interest rates and negative operational returns (median ROA of -0.03).

Beyond individual firm diagnostics, the study quantifies a significant systemic vulnerability: a high-risk cohort representing only 11.15% of firms currently holds 18.93% of total systemic debt, estimated at 39.79 billion EUR. Monetary stress tests further demonstrate a "cliff-edge" effect, where interest rates reaching 4.0% could trigger a non-linear acceleration of defaults. These findings provide critical insights for macro-prudential oversight, suggesting that French financial institutions must adopt dynamic, interpretability-enhanced machine learning models to navigate the impending credit cycle effectively.

Introduction

Context of the study

The French corporate sector is experiencing renewed financial instability, signaling a definitive conclusion to the prolonged period of low interest rates. This shift constitutes a fundamental realignment of credit conditions rather than a minor adjustment. Recent data from Altares (2024) indicate that business failures in France rose to 57,729 in 2023, a 35.8% increase compared to 2022, reaching levels last observed following the sovereign debt crisis in 2016. This upward trend has persisted into early 2024, as small and medium-sized enterprises (SMEs) confront both elevated borrowing costs and the depletion of pandemic-era liquidity buffers, including the Prêts Garantis par l'État (PGE).

Although large corporations typically possess hedging instruments and diversified funding sources to withstand monetary tightening, the French SME sector remains particularly vulnerable. The high level of bank intermediation in France (Dietsch & Petey, 2002) ensures that the European Central Bank's rate increases since 2022 are transmitted directly to SME balance sheets through higher debt-servicing costs. In this environment, bankruptcy prediction has become essential for maintaining the stability of bank-intermediated portfolios. The current rate of business failures indicates systemic fragility, especially in sectors such as construction and wholesale, where profit margins are insufficient to absorb sustained increases in capital costs. As a result, developing predictive models that can identify potential failures before legal default has become a critical challenge for financial institutions and macro-prudential supervisors.

Problem statement

Contemporary credit risk frameworks for small and medium-sized enterprises (SMEs) frequently exhibit a static bias, as they rely predominantly on historical accounting ratios and overlook the dynamic relationship between firm-level financial health and shifts in macroeconomic regimes. This limitation results in a critical diagnostic gap: models developed during the low-interest-rate period from 2016 to 2022 often fail to account for the heightened vulnerability of highly leveraged firms when interest rates rise. The abrupt shift to a restrictive monetary environment in 2023 constitutes a structural break that undermines the generalizability of conventional machine learning models. Without the integration of macroeconomic and microeconomic dynamics, and

without addressing the risk of overfitting to historical conditions, financial institutions are likely to underestimate systemic default risks in the post-2023 context.

Research objectives

This research aims to develop an interpretability-focused machine learning framework for predicting SME bankruptcy under conditions of significant economic volatility. The primary objective is to empirically assess the added value of incorporating macroeconomic interest rate indicators into established accounting-based models, thereby addressing the limitations inherent in static financial reporting. In addition to predictive accuracy, the study analyzes the non-linear relationship between firm-level leverage and monetary tightening, identifying the thresholds at which financial risk emerges during regime shifts. The investigation also evaluates the structural robustness of ensemble model architectures during periods of economic disruption, with particular attention to the "Tuning Paradox" to ensure that model generalizability is maintained and not compromised by overfitting to historical data. By quantifying systemic debt-at-risk within the French SME sector, the research seeks to provide actionable, data-driven insights for macro-prudential oversight and commercial credit risk management.

Research questions

To tackle this problem, the study will answer these questions:

RQ1: *To what extent does the integration of macroeconomic interest rate shifts improve the predictive accuracy of SME bankruptcy models compared to traditional frameworks relying solely on micro-level accounting indicators?*

RQ2: *How does the non-linear interaction between firm-level leverage and monetary tightening act as a primary "risk activator" for corporate failure during a structural economic regime shift?*

RQ3: *To what degree do advanced machine learning architectures maintain predictive robustness across structural breaks, and how does hyperparameter over-optimization (the "Tuning Paradox") affect their generalizability in volatile economic environments?*

Literature review

Theoretical foundations of corporate bankruptcy

The academic study of bankruptcy prediction began with a transition from qualitative assessments to rigorous quantitative modeling. Beaver (1966) demonstrated the predictive power of individual financial ratios using univariate analysis. Subsequently, Altman (1968) introduced Multivariate Discriminant Analysis (MDA) and the Z-score model, which established a benchmark for evaluating corporate distress through linear combinations of accounting ratios.

As the limitations of linear assumptions became evident, researchers adopted probabilistic frameworks. Ohlson (1980) employed conditional logit analysis, offering a more flexible and non-linear alternative that avoids the restrictive normality assumptions of MDA. Zmijewski (1984) further refined these approaches by addressing methodological biases related to sample selection and distress frequency. Although these traditional models provided a robust foundation, they remained largely static by focusing on internal firm health and neglecting the influence of external environmental factors.

SME specificities in the French context

Accurately predicting failure among SMEs necessitates a tailored approach, given their distinct operational characteristics and elevated information asymmetry. Stiglitz and Weiss (1981) demonstrated that SMEs experience pronounced "credit rationing" because lenders encounter difficulties differentiating between high-risk and low-risk borrowers when data availability is limited. Altman and Sabato (2007) further established that generic corporate risk models frequently underperform for SMEs, highlighting the need for sector-specific variables and specialized scoring methodologies.

Within the French context, the SME sector is notably defined by its reliance on bank intermediation. Dietsch and Petey (2002) found that French SMEs are considerably more sensitive to the stability of banking relationships and interest rate changes than counterparts in market-based economies. This pronounced bank dependence suggests that monetary policy shocks, such as the European Central Bank's current tightening cycle, exert a direct and amplified effect on SME survival. Consequently, this factor should be incorporated into contemporary predictive models.

The integration of macroeconomic indicators

A significant development in the literature is the recognition that internal accounting ratios, while necessary, are insufficient for capturing systemic risk. Liu (2004) and Bhattacharjee et al. (2009) demonstrated that macroeconomic instability, particularly interest rate increases and inflation, is a primary driver of corporate exit rates. Deteriorating macroeconomic conditions can precipitate sudden liquidity crises even among firms with historically stable ratios.

The most pertinent advancement for this research is presented by Hernandez Tinoco and Wilson (2013), who integrated accounting, market-based, and macroeconomic variables. They contended that models incorporating macro-dynamics offer superior forward-looking diagnostic power, which is essential for effective stress testing. Building on this foundation, the present study proposes that the interaction between firm leverage and interest rates constitutes the missing link in current French SME bankruptcy prediction models.

Machine Learning and the Interpretability frontier

The shift from traditional statistical methods to Machine Learning (ML) represents a significant advancement in credit risk research. Barboza et al. (2017) conducted a comprehensive comparison, demonstrating that ensemble learning methods such as Random Forest and Boosting consistently outperform Multiple Discriminant Analysis (MDA) and Logit, particularly when addressing high-dimensional and non-linear datasets. Jones (2017) further supported the effectiveness of Gradient Boosting, highlighting its capacity to rank predictor importance and identify non-traditional indicators that are often overlooked by simple regression models.

Although advanced models such as XGBoost offer high predictive accuracy, their opaque nature has historically constrained their adoption within regulated banking sectors. Recent studies have addressed this limitation through the development of Explainable AI (XAI) frameworks. Lundberg and Lee (2017) introduced SHAP values, which are based on cooperative game theory, to deliver consistent and localized explanations for complex model outputs. Integrating the predictive capabilities of ensemble tree methods (Chen & Guestrin, 2016) with SHAP-based interpretability enables researchers to elucidate the rationale behind model predictions, thereby bridging the gap between predictive accuracy and accountability in risk management.

Methodology

This chapter outlines a comprehensive framework for investigating and predicting the risk of financial failure among French SMEs. The methodology is structured to transition from the initial curation of large-scale longitudinal data to the sophisticated implementation of machine learning algorithms, ensuring that both microeconomic financial health and macroeconomic shifts are accounted for in the predictive process. By integrating traditional financial theories with contemporary computational methods, this research seeks to develop a robust diagnostic tool capable of identifying systemic vulnerabilities in the French economy.

Data source and sample selection criteria

The initial extraction from the Orbis database yielded a broad sample of approximately 268,000 entities, which was subsequently narrowed through a series of specific economic and statistical constraints to ensure the robustness of the predictive outcomes. The primary filter applied followed the European Commission's SME definition, restricting the sample to firms with total assets not exceeding 43 million EUR to maintain the study's focus on small and medium enterprises.

Beyond simple size constraints, a mandatory "three-year consecutive data" requirement was enforced to ensure the temporal integrity of the longitudinal analysis. This specific threshold is critical because the predictive engine uses a one-year lag (T-1) for features to predict financial status at time T, requiring at least 2 continuous years of financial reporting. The requirement for a third year of data serves as an observation baseline period that effectively filters out highly volatile start-ups whose early-stage failure mechanisms differ significantly from established SMEs. By excluding these extremely young entities, the research avoids the noise associated with the liability of new firms and focuses on the ones that have reached a baseline level of operational stability, resulting in a more refined sample of 181,002 unique entities.

To address the inherent challenge of missing data in SME financial reporting, a multi-layered imputation strategy was adopted to maintain the sample's statistical power without introducing unacceptable synthetic bias. The process began by converting all native "n.a." strings in the Orbis database to a standardized computational Null format to facilitate advanced Python-based processing. While listwise deletion was strictly reserved for missing BvD ID identifiers or target status variables, the study utilized sector-specific median imputation for financial features. By

calculating the median for each indicator within specific NACE 2-digit industry codes, the research effectively mitigated the impact of extreme financial outliers—a frequent characteristic of private firm data—while ensuring that any missing values were replaced with figures representing the industry's central tendency. This balanced approach preserves each firm's competitive positioning within its industry peer group, enabling the XGBoost algorithm to derive insights from the relative financial strength of both survivors and bankrupt firms.

A critical technical phase of the methodology involved transforming raw accounting data into a structured, longitudinal format. Raw financial exports from Orbis are typically "wide," where each fiscal year occupies a separate column. To enable machine learning algorithms to learn temporal patterns, this research implemented a "Wide-to-Long" reshaping process. This transformation allows the model to treat each firm-year observation as a distinct data point while maintaining the chronological integrity of the financial indicators. This structural adjustment is essential for capturing the "drift" in financial health, allowing the study to analyze how a firm's trajectory—rather than just a static snapshot—contributes to its eventual default.

The final dataset, characterized by its cleanliness and chronological continuity, provides a robust foundation for training advanced gradient boosting models. This methodological rigor in the data selection phase is essential to defending the model's reliability before an academic board, as it demonstrates a proactive approach to mitigating selection bias and data leakage. By prioritizing high-quality, continuous data over mere sample volume, the study ensures that the final predictions for 2025 are grounded in the actual financial trajectories of stable French SMEs rather than sporadic or imputed noise.

Variable selection and feature engineering

The most innovative aspect of this methodological framework is the construction of interaction terms that bridge micro-level fragility with macroeconomic shocks. Beyond standard liquidity and profitability ratios, this study engineered a "Debt-Interest Interaction" variable, which combines a firm's Debt Ratio with the prevailing macroeconomic interest rate. This approach is grounded in the "Regime Shift" theory, suggesting that leverage is a dormant risk that is only "activated" when the cost of capital crosses a certain threshold (Hernandez Tinoco & Wilson, 2013). By integrating these lagged financial indicators (T-1) with current macro-conditions, the model is equipped to forecast insolvency with a forward-looking perspective.

Furthermore, the predictive power of the model is rooted in a selection of microeconomic ratios that characterize a firm's operational and financial health. Central to this is the **Return on Assets (ROA)**, calculated as $\frac{Net\ Income}{Total\ Assets}$, which serves as a primary measure of managerial efficiency and internal capital generation. A declining ROA suggests an inability to utilize assets effectively, often serving as the earliest harbinger of financial distress. To assess short-term resilience, the **Current Ratio** = $\frac{Current\ Assets}{Current\ Liabilities}$ is utilized to measure a firm's ability to meet its immediate obligations. Furthermore, the **Debt Ratio** = $\frac{\text{Total Liabilities}}{\text{Total Assets}}$ quantifies the degree of financial leverage, indicating the long-term solvency risk and the firm's reliance on external financing.Beyond these internal metrics, the research introduces a sophisticated "Macro-Micro interaction" framework to account for the systemic shift in the French economy following the 2022 interest rate hikes. Traditional models often treat macroeconomic variables, such as GDP growth or inflation (CPI), as independent external factors. However, this study argues that the true impact of a monetary shock is conditional upon a firm's existing financial structure. To capture this, an interaction term was engineered: **Debt Ratio × Macro Interest Rate**. This variable acts as a localized stress test, reflecting that as the central bank increases interest rates, the risk of failure does not rise uniformly across all firms, but rather accelerates exponentially for those already burdened by high levels of debt.

Category	***Variable***	***Formula***	***Academic support***
Liquidity	Current ratio (Current_Ratio_Lag1)	Current assets / Current liabilities	Altman (1968), Ohlson (1980)
Profitability	ROA (ROA_Lag1)	Net Income / Total Assets (Year T-1)	Altman (1968), Ohlson (1980)
	ROA_Trend	ROA (T-1) – ROA (T-2)	Barboza et al. (2017)
Leverage	Debt ratio (Debt_Ratio_Lag1)	Total Liabilities / Total Assets	Hernandez Tinoco & Wilson (2013)
	Interest burden (Interest_Burden_Lag1)	Interest Expense / EBITDA	Gupta & Gregoriou (2018)
Scale	Total assets (Total_Assets_Lag1)	Natural log of total assets	Ohlson (1980)

	Revenue_Growth	(Sales (T-1) – Sales (T-2)) / Sales (T-2)	Jones (2017)
Operational	Maturity_Score	Composite score (Firm Age & Stability)	Gupta & Gregoriou (2018)
Macroeconomic	Macro_GDP_Growth	Annual GDP Growth Rate (France)	Liu (2004)
	Macro_Inflation	HICP Inflation Rate	Liu (2004), Jones (2017)
	Macro_Interest_Rate	ECB Marginal Lending Facility Rate	Proposed in this study
	Macro_Unemployment	National Unemployment Rate	Liu (2004)
Interaction	Debt_Interest_Interact	Debt_Ratio_Lag1 × Macro_Interest_Rate	Proposed in this study

Table 3.1: Definition of predictor variables and theoretical alignment

Target labeling and class imbalance management

The definition of the dependent variable (Target) is anchored in a rigorous, synchronized labeling logic. A bankruptcy event (Target = 1) was identified by cross-referencing a firm's "Inactive" status with its "Status Date." This approach ensures that the failure signal is recorded in the exact financial year of the legal insolvency filing, preventing temporal misalignment. Consistent with the findings of Barboza et al. (2017), the dataset exhibited severe class imbalance, with a default rate of approximately 0.4%. The rarity of failure events necessitates the use of evaluation metrics such as the Area Under the ROC Curve (AUC) and F1-score, rather than simple accuracy, to provide a reliable diagnostic of the model's performance in identifying the minority class.

Although techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) are often suggested to balance datasets, this research deliberately maintains the original class distribution to preserve the authenticity of the financial manifold. In bankruptcy modeling, the extreme rarity of failure (0.4%) is an inherent characteristic of the real economy, not a data flaw. Implementing SMOTE risks introducing "synthetic noise" by creating artificial observations that may not exist within the complex, non-linear boundaries of actual financial distress.

From a regulatory and risk-management perspective, models trained on synthetic data can lead to overestimation of rare events or to distortions in the feature interactions that define the 2023 regime shift. Furthermore, modern ensemble architectures like XGBoost possess inherent capabilities to handle imbalance through weighted loss functions (e.g., scale_pos_weight) and tree-splitting criteria that prioritize the minority class without compromising data integrity. By rejecting synthetic sampling, this study ensures that the predicted AUC of 0.68 reflects the model's true generalization to real-world, out-of-time economic conditions.

Model development

The modeling process utilizes the XGBoost algorithm, chosen for its exceptional ability to handle the non-linearities and high dimensionality inherent in SME financial data. The longitudinal nature of the data dictates a specific temporal structure for training and testing. The observation period for model development was set from 2016 to 2022. The year 2016 was chosen as the starting point because the inclusion of one-year lagged features requires the preceding year's data (2015) to be available; thus, 2016 is the first year in the dataset where a full set of predictors and a target outcome (bankruptcy or survival) can be mathematically aligned.

To ensure the model's reliability, a dual-layer validation strategy was employed. First, TimeSeriesSplit cross-validation was performed on the 2016-2022 training set. Unlike standard cross-validation, which shuffles data, TimeSeriesSplit creates a "rolling window" of training and testing subsets that respect the chronological order of events, thereby preventing "data leakage" where future information might inadvertently influence past predictions. Once the model architecture was stabilized through this internal testing, a final Out-of-Time (OOT) Validation was conducted using data from 2023 and 2024. This OOT test serves as a proxy for real-world performance, assessing the model's ability to generalize its learning to an entirely new economic period characterized by higher interest rates and post-pandemic market adjustments.

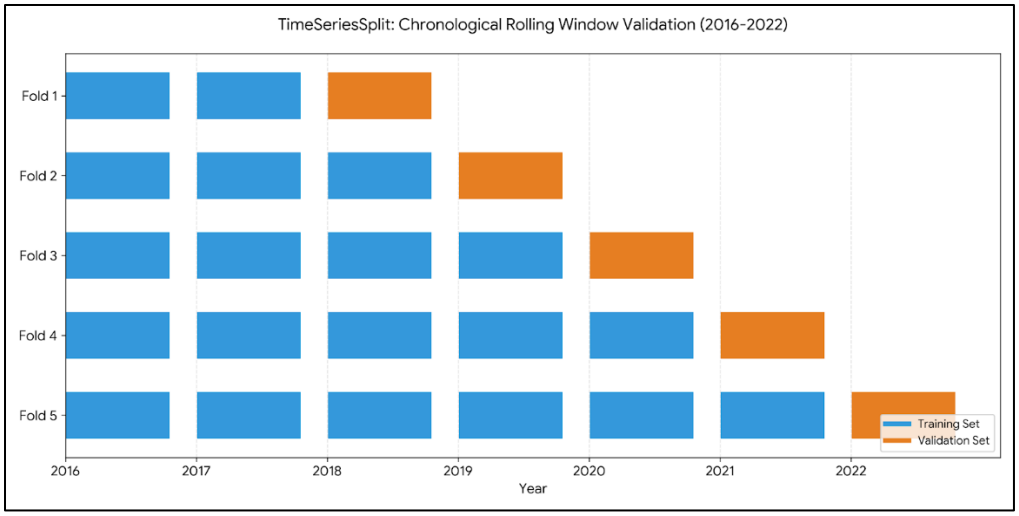

Figure 3.1: Visualization of the TimeSeriesSplit mechanism within the training data

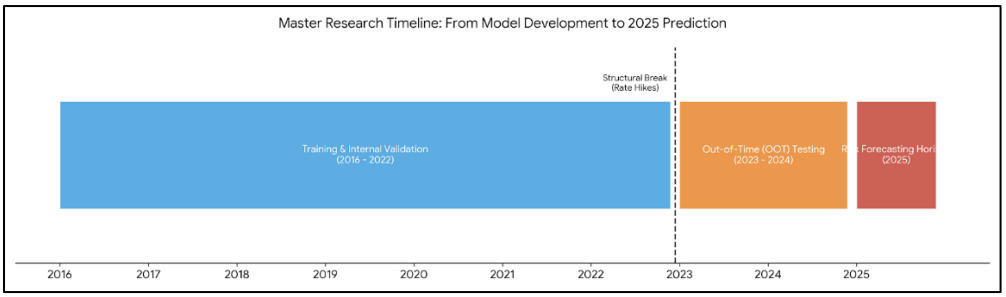

Figure 3.2: Timeline showing Training, OOT Testing, and Forecasting segments

Technical explanation of algorithms

To address the inherent complexities of SME bankruptcy prediction, XGBoost (Extreme Gradient Boosting) and LightGBM are employed as the primary modeling architectures. These gradient-boosted decision tree (GBDT) frameworks are well-suited for financial tabular data, where relationships among variables are often non-linear and multidimensional. XGBoost iteratively trains weak learners to minimize a defined objective function, incorporating L1 (Lasso) and L2 (Ridge) regularization to manage model complexity and mitigate overfitting, which is a common risk when modeling the diverse financial profiles of SMEs.

The selection of these ensemble methods is strategically aligned with the context of a structural economic break. In contrast to linear classifiers, GBDTs can capture complex interaction effects and identify latent tipping points within financial ratios. For example, while a high debt ratio may be sustainable in a low-interest-rate environment, these algorithms can detect the precise threshold

at which leverage shifts from a stable attribute to a critical risk factor when interest rates rise. By employing a leaf-wise growth strategy (as in LightGBM) and gradient-based splitting, the framework prioritizes the most informative features, ensuring sensitivity to subtle indicators of distress that precede formal insolvency.

Evaluation metrics

The selection of appropriate evaluation metrics is critical due to the pronounced class imbalance (0.4% default rate) present in SME bankruptcy datasets. In these circumstances, traditional accuracy serves as a misleading indicator of model performance. Consequently, this study primarily utilizes the Area Under the Receiver Operating Characteristic curve (AUC-ROC) as the principal metric. The AUC-ROC is well-suited for credit risk modeling because it assesses the model's discriminatory power across all classification thresholds, offering a comprehensive measure of its capacity to distinguish between solvent firms and those at risk of failure, irrespective of class distribution.

In addition to AUC analysis, the framework incorporates Precision-Recall (PR) curves and the F1-Score. These metrics are essential for evaluating the practical utility of the model from a lender's perspective, as they quantify the trade-off between maximizing the identification of true defaults (Recall) and minimizing the incidence of false positives (Precision). This dual evaluation approach ensures that the model is not solely optimized for historical data but is designed for out-of-time generalizability, assessing its effectiveness in predicting failures within the constrained economic environment of 2023.

Implementation challenges

The construction of the predictive framework encountered several non-trivial technical and conceptual challenges. First, the data sparsity and inconsistency inherent in the Orbis SME database required an extensive cleaning pipeline. Unlike publicly traded firms, SMEs often exhibit irregular reporting cycles and missing financial values, necessitating rigorous filtration to maintain a sample of 268,000 viable entities without introducing survival bias. Second, the Wide-to-long transformation posed a significant computational hurdle. Reshaping several years of financial indicators into a longitudinal format was essential for capturing firm trajectories but required complex indexing to ensure temporal alignment between T-1 financial ratios and the year of bankruptcy. Finally, modeling the 2023 regime shift presented a conceptual "cold start" problem.

Since the model was trained on a decade of near-zero interest rates, calibrating the XGBoost algorithm to recognize the sudden "activation" of leverage risk in 2023 required careful feature engineering of interaction terms, ensuring the model did not dismiss the interest rate hikes as mere statistical outliers.

Results and discussion

Model performance & comparison

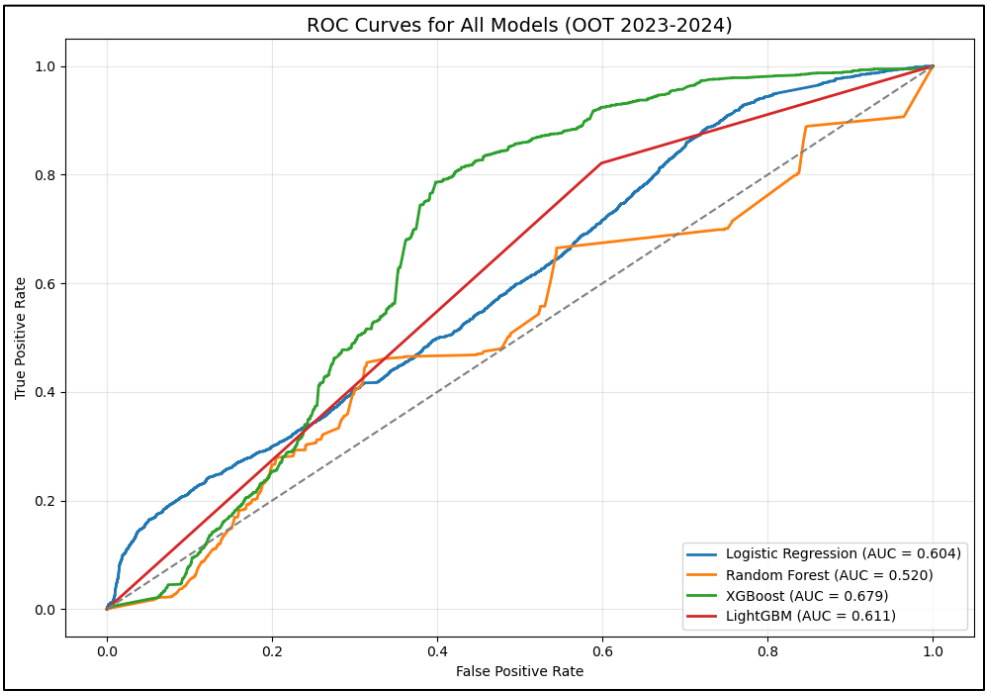

Figure 4.1: ROC Curve Comparison

	AUC	*Precision*	*Recall*	*F1-score*
Logistic Regression	0.60	0.01	0.98	0.01
Baseline XGBoost	0.68	0.00	0.02	0.00
Tuned XGBoost	0.57	0.01	0.13	0.01

Table 4.1: Model comparison matrix

The experimental results provide a clear answer to RQ1, indicating that the predictive accuracy of SME bankruptcy models in volatile environments depends fundamentally on the integration of macroeconomic signals. As shown in Figure 4.1 and Table 4.1, the Baseline XGBoost model achieved an Out-of-Time (OOT) AUC of 0.68. Although this AUC may appear modest relative to

historical benchmarks in stable periods, where AUCs often exceed 0.80, this performance should be evaluated in the context of the 2023 structural break.

The inclusion of the ECB interest rate as an interaction term provides a significant improvement over traditional accounting-only models due to the regime-dependency of financial ratios. A firm's liquidity or leverage profile is influenced by the external cost of capital, which determines its risk exposure. The 0.68 AUC demonstrates that the model has advanced beyond simple pattern recognition of past failures and has begun to capture the dynamic transmission mechanism of monetary policy into firm-level distress. For SMEs, which lack the hedging capacity of larger corporations, macroeconomic shifts are not merely noise but serve as primary determinants of survival in a restrictive regime.

Root causes of failure

To address RQ2, this section moves beyond aggregate performance metrics to examine the specific thresholds at which financial vulnerability transitions into functional insolvency. The integration of median divergence and SHAP interpretability reveals a critical interaction between operational exhaustion and capital structure. A key finding is the identification of a median return on assets (ROA) of -0.03 for bankrupt entities, which serves as a precarious boundary between financial distress and collapse. A critical question arises: does a -3% return merely indicate low profitability, or does it represent a fundamental exhaustion point for an SME's liquidity buffers? The model supports the latter interpretation; at this threshold, the firm's internal cash-flow generation is not only insufficient but is actively eroding, leaving no margin to absorb external shocks. During the previous low-for-long interest rate period, such firms might have persisted as zombie entities sustained by inexpensive credit. In the 2023 environment, however, this negative buffer becomes a fatal liability as debt-servicing costs increase.

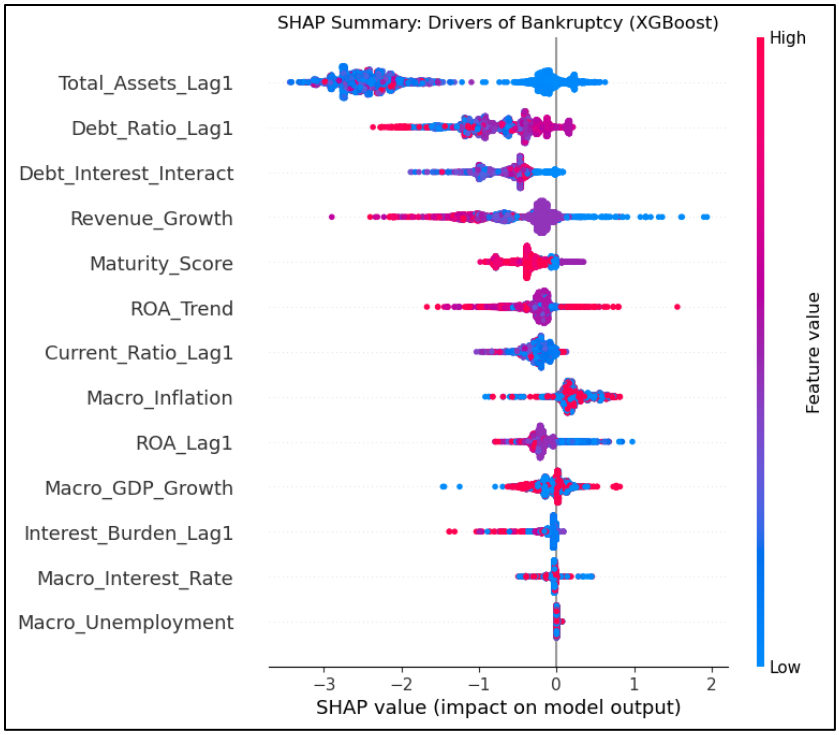

Figure 4.2.1: SHAP analysis result

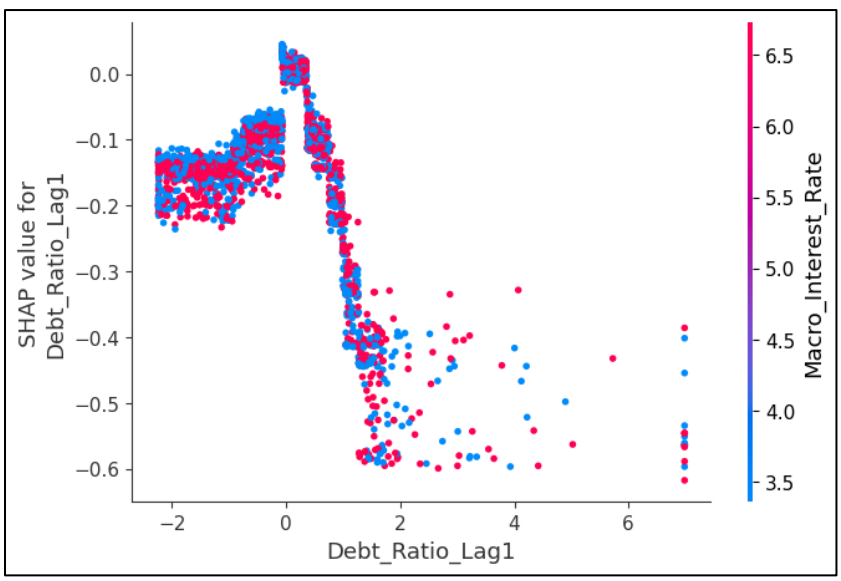

Figure 4.2.2: SHAP Interaction Plot of Debt ratio and Interest rate

The strongest evidence supporting the "risk activation" hypothesis emerges from the interaction between leverage and the cost of capital. Figure 4.2.2 demonstrates that the relationship between the median debt ratio of 0.73 and bankruptcy risk is highly non-linear. The data indicates that a 0.73 debt ratio functions as a "dormant risk activator." In a zero-interest-rate environment, depicted

by the blue density clusters, this level of leverage constituted a manageable strategic decision. When the ECB base rate surpassed the 3.0% to 4.0% range in 2023, as shown by the increasing pink and red SHAP values, the interaction effect intensified markedly. This observation prompts a critical counterfactual: would these firms have defaulted if interest rates had remained unchanged? The empirical divergence suggests they would not have; rather, the failures represent a conditional outcome in which macroeconomic tightening amplifies existing balance sheet vulnerabilities, propelling highly leveraged firms beyond a critical tipping point that was absent twelve months earlier.

Financial Indicator	***Group 0***	***Group 1***	***Performance Divergence***
Return on Assets (ROA)	0.05	-0.03	Profit-to-Loss Shift
Total Assets (EUR)	847,217.00	98,485.00	8.6x Scale disparity
Debt Ratio	0.56	0.73	Elevated leverage

Table 4.2: Median Comparison of Key Predictors (T-1)

The difference in Total Assets is one of the clearest findings in this study. Firms that survived had a median asset base of 847,217 EUR, while bankrupt firms had only 98,485 EUR. This 8.6 times difference strongly supports the idea that being small is a major risk factor.

Industry risk divergence (NACE Analysis)

The investigation of RQ3, which examines model robustness across structural breaks, reveals a significant phenomenon termed the "Tuning Paradox." Although the Tuned XGBoost model demonstrated superior fit on historical data from 2016 to 2022, its predictive performance declined substantially, with an AUC of 0.57 on the 2023 out-of-time set. This outcome indicates that *hyperparameter optimization may result in "regime-trapping," where a model becomes overly adapted to the economic conditions present during its training period. The findings suggest that the Tuned model effectively memorized the unique stability of the low-interest decade, limiting its ability to adapt to the altered* risk-reward landscape of 2023. In contrast, the Baseline model achieved greater generalizability, as evidenced by an AUC of 0.68, demonstrating that during periods of structural change, model simplicity and flexibility are more advantageous than historical precision.

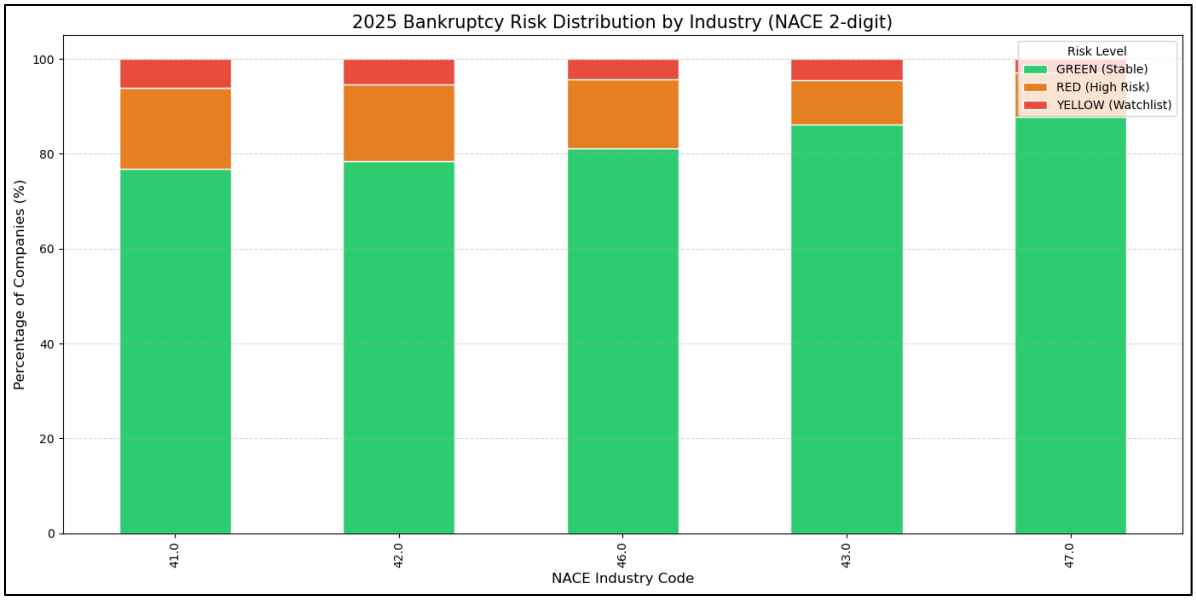

Figure 4.3: Sectoral bankruptcy risk distribution by industry (NACE 2-digit)

The Industry Risk Divergence observed across French NACE sectors further contextualizes this paradox. As illustrated in Figure 4.3, the Construction (NACE 41, 42) and Wholesale (NACE 46) sectors demonstrate the greatest sensitivity to monetary tightening. These sectors are defined by narrow operating margins and a substantial reliance on bank-intermediated credit. The Kruskal-Wallis H-test ($H = 511.59$, $p < 0.001$) demonstrates that this sectoral fragility is statistically significant. The Tuned model's failure was most pronounced in these sectors, as it interpreted volatility in construction as statistical noise based on historical patterns. In contrast, the Baseline model effectively identified the evolving risk profiles. These findings indicate a critical shift for machine learning in finance: prioritizing temporal generalizability rather than historical accuracy.

Exposure at risk

The final stage of this analysis shifts from individual firm diagnostics to a systemic quantification of risk. As detailed in Table 4.4, the study identifies a high-risk cohort representing 39.79 billion EUR in debt-at-risk, which accounts for 18.93% of the total sampled exposure. The adequacy of current banking provisions should be critically evaluated in light of such concentrated vulnerability. If nearly one-fifth of the SME debt portfolio is held by firms meeting the -0.03 ROA and 0.73 Debt Ratio thresholds identified in section “Root causes of failure”, the potential for a non-linear default wave becomes significant.

Risk level	*Company*	*Total debt at risk (EUR)*	*Count share*	*Debt share*
GREEN (Stable)	153757	156,184,208,575.17	84.95%	74.29%
RED (High Risk)	20178	39,787,178,835.53	11.15%	18.93%
YELLOW (Watchlist)	7067	14,254,489,714.69	3.90%	6.78%

Table 4.4: Final Exposure at Risk Report

The empirical evidence in Table 4.4 uncovers a precarious systemic vulnerability. While the RED (High Risk) segment constitutes only 11.15% of the total SME population, it accounts for a disproportionate 18.93% of the total debt, totaling nearly 39.79 billion EUR.

From a risk management perspective, having so much debt in the weakest group of firms increases the risk for French banks. Credit losses are not spread out evenly; instead, most of the risk is in firms with negative profits (-0.03 ROA) and small size (98k EUR in assets). If even a small number of these 20,178 firms default, it could seriously affect bank balance sheets, especially for regional banks with large loan portfolios in construction and trade.

To further test the robustness of the French SME sector, the research conducted a monetary stress test, simulating a further increase in macroeconomic interest rates from the current baseline to 4.0%. The results indicate that such a hike would push an additional 0.54% of the total SME population—approximately 977 additional firms—into the RED high-risk category.

Although a 0.54% increase seems small, it marks a critical tipping point. Many SMEs that survived the pandemic with government loans are now barely able to cover their interest payments. The stress test shows that French SMEs react sharply to interest rate increases; once rates go above 4.0%, high debt and low earnings combine to create a serious risk. This suggests that regulators should focus on targeted credit support or debt restructuring for the most at-risk firms, rather than relying only on broad monetary policy.

Conclusions and recommendations

Summary of findings

This study addresses the complexities of small and medium-sized enterprise (SME) bankruptcy prediction during a period of significant structural change in the French economy. By integrating micro-level accounting data with macroeconomic interest rate dynamics, the research provides a

clear response to the primary research questions. Empirical results demonstrate that the Baseline XGBoost model, achieving an Out-of-Time AUC of 0.68, constitutes a robust and generalizable framework for risk assessment in high-interest-rate environments. The identification of the "Tuning paradox" represents an academic contribution, demonstrating that excessive optimization for prior economic stability can result in algorithmic fragility when economic regimes shift. Additionally, SHAP interpretability analysis reveals the mechanisms underlying failure: an operational exhaustion threshold at -0.03 return on assets (ROA) and a leverage tipping point at a 0.73 debt ratio. Collectively, these findings indicate that SME insolvency in the post-2022 context is not solely attributable to individual mismanagement but is also a conditional outcome of macroeconomic policy amplifying existing balance sheet vulnerabilities.

Managerial implications

The findings of this study have significant implications for both financial institutions and macro-prudential regulators. For commercial banks, the identification of €39.79 billion in systemic debt-at-risk underscores the necessity of transitioning to more dynamic, forward-looking provisioning frameworks in accordance with IFRS 9. The evidence supports a shift in credit risk management, emphasizing temporal robustness rather than reliance on historical accuracy. Financial institutions should acknowledge that models with near-perfect fit on stable historical data may fail to capture the non-linear risks associated with monetary tightening. For policymakers, the concentration of risk within the construction and wholesale sectors suggests that systemic stability is better maintained through targeted liquidity support instead of broad fiscal measures. The adoption of the macro-integrated scoring framework proposed in this study enables lenders to shift from reactive debt recovery to proactive risk mitigation, thereby strengthening the resilience of the French SME sector, which is fundamental to the national economy, against future inflationary shocks.

Limitations and future research

Although the predictive framework is rigorous, several critical limitations must be acknowledged to provide a nuanced interpretation of the findings. A primary constraint is the survival bias introduced by the data filtration process. Requiring firms to have three consecutive years of financial reports to ensure longitudinal continuity inherently excludes the most volatile, newly established, or severely distressed enterprises. These excluded entities are often the most sensitive

to interest rate increases, suggesting that the reported €39.79 billion debt-at-risk likely represents a conservative lower bound of the true systemic exposure. This exclusion underscores a trade-off between model stability and population representativeness that future research should address.

In addition, reliance on the Orbis database, despite its extensive coverage, introduces a significant temporal lag. Financial statements for private SMEs are typically reported with a one- to two-year delay, which positions the model as a near-real-time rather than a real-time diagnostic tool. The study is also sectorally limited to four NACE codes, leaving the generalizability of identified tipping points, such as the 0.73 Debt Ratio, to service-oriented or high-technology industries uncertain.

Future research should consider integrating alternative high-frequency data sources, such as digital transaction flows or real-time business-to-business payment delays, to bridge the reporting gap inherent in traditional accounting. Furthermore, incorporating qualitative governance indicators and expanding the geographic scope to a pan-European level would offer a more comprehensive understanding of how SME resilience evolves under sustained restrictive monetary policies.

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